

A Minor Project Report On
DRINKING WATER QUALITY CHECKER



Submitted in the Partial Fulfillment of the
Requirements for the Degree of Bachelor of Computer Engineering
Awarded by Pokhara University

Submitted By:

Abhishek Acharya (22070123)

Pankaj Kumar Mandal (22070137)

Rajendra Bahadur Pachhai (22070143)

Sadip Bahadur Thapa (22070147)

Supervisor:

Er. Sushant Paudel

Assistant Professor

School of Engineering
Faculty of Science and Technology
POKHARA UNIVERSITY

August, 2025

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Er. Bhesh Bahadur Thapa
Program Coordinator
School of Engineering
Pokhara University

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We hereby declare that this project work entitled **DRINKING WATER QUALITY CHECKER** is based on our original work. All concepts, data, code, and any other work from external sources have been properly cited and referenced in accordance with the guidelines provided by School of Engineering, Pokhara University

We owe all the liabilities relating to the authenticity and originality of this project work and project report.

1. Abhishek Acharya (22070123)
2. Pankaj Kumar Mandal (22070137)
3. Rajendra Bahadur Pachhai (22070143)
4. Sadip Bahadur Thapa (22070147)

Date:

SUPERVISOR'S RECOMMENDATION

This is to certify that this project report entitled **DRINKING WATER QUALITY CHECKER** prepared and submitted by the below listed team of students in partial fulfillment of the requirements of the degree of Bachelor of Computer Engineering awarded by Pokhara University, has been prepared and completed under my supervision.

I hereby recommend the same for acceptance by the School of Engineering, Pokhara University.

1. Abhishek Acharya (22070123)
2. Pankaj Kumar Mandal (22070137)
3. Rajendra Bahadur Pachhai (22070143)
4. Sadip Bahadur Thapa (22070147)

.....

Er. Sushant Paudel

Assistant Professor

School of Engineering

Pokhara University

Date:



School of Engineering POKHARA UNIVERSITY

EXTERNAL EXAMINER'S RECOMMENDATION

The undersigned certified that they have evaluated this project report entitled **DRINKING WATER QUALITY CHECKER** submitted by **Abhishek Acharya, Pankaj Kumar Mandal, Rajendra Bahadur Pachhai, Sadip Bahadur Thapa** and their oral presentation for partial fulfillment of the degree of Bachelor of Computer Engineering and recommended to the School of Engineering, Pokhara University for acceptance of this project work/report.

.....

Er. Raj Kiran Chhatkuli

Assistant Professor

IOE Paschimanchal Campus

Lamachaur, Pokhara, Kaski

Date:



School of Engineering POKHARA UNIVERSITY

LETTER OF APPROVAL

This project report entitled **DRINKING WATER QUALITY CHECKER** submitted by **Abhishek Acharya, Pankaj Kumar Mandal, Rajendra Bahadur Pachhai, Sadip Bahadur Thapa** for partial fulfillment of the degree of Bachelor of Computer Engineering has been accepted by the School of Engineering, Pokhara University upon the recommendations of Supervisor and with the approval by the following examiner.

Er. Raj Kiran Chhatkuli

Assistant Professor

IOE Paschimanchal Campus

Lamachaur, Pokhara, Kaski

Er. Bhesh Bahadur Thapa

Assistant Professor

(Coordinator)

Bachelor of Computer Engineering

Bachelor of Software Engineering

School of Engineering

Pokhara University

Date:

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to Assist. Prof. Er. Sushant Paudel for his valuable guidance, encouragement, and continuous support throughout the development of our project, Learned Image Compression Using Autoencoder. We are also thankful to Pokhara University for providing us with the opportunity and environment to explore, research, and innovate in the field of deep learning and image processing.

We would like to acknowledge the contributions of open-source platforms, public datasets, and detailed documentation that played a vital role in building and training our machine learning models. We are also grateful to the online developer and research communities whose insights and discussions helped us overcome various technical challenges.

This project would not have been possible without the collaboration and effort of our entire team. We are grateful for the teamwork, dedication, and learning we experienced together throughout the journey.

Abhishek Acharya	[22070123]
Pankaj Kumar Mandal	[22070137]
Rajendra Bahadur Pachhai	[22070143]
Sadip Bahadur Thapa	[22070147]

ABSTRACT

This project centers on creating a low-cost, user-friendly system for monitoring drinking water quality using an Arduino microcontroller and a TDS (Total Dissolved Solids) sensor. The device measures TDS levels, which reflect the concentration of dissolved particles in water, and displays this data on an LCD screen. Based on standard water quality benchmarks, the system classifies water into categories like "Excellent," "Good," "Fair," or "Poor," allowing users to quickly assess its safety for consumption.

The primary goal is to develop a portable and accessible solution for real-time water quality assessment suitable for domestic, industrial, and environmental use. By offering immediate feedback, this system helps ensure the availability of safe drinking water and can play a critical role in promoting public health and environmental monitoring efforts.

Keywords: TDS sensor, Arduino Microcontroller, Real-Time Water Quality checker, Android application.

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LIST OF ABBREVIATIONS

IDE	Integrated Development Environment
LCD	Liquid Crystal Display
OS	Operating System
PPM	Parts Per Million
RO	Reverse Osmosis
SDLC	System Development Life Cycle
TDS	Total Dissolved Solid
USB	Universal Serial Bus
Wi-Fi	Wireless Fidelity

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|CHAPTER 1

INTRODUCTION

1.1 Background

Water quality has been a critical concern for centuries, with various civilizations developing methods to ensure safe drinking water. In modern times, water quality is determined using multiple parameters, including pH, temperature, turbidity, and Total Dissolved Solids (TDS). Among these, TDS is a key indicator of water purity, representing the combined content of all inorganic and organic substances dissolved in water.

High TDS levels can indicate the presence of harmful substances, such as heavy metals, salts, and other pollutants, making it an essential parameter for water quality testing. Conventional methods for water quality assessment are often expensive, time-consuming, and require specialized equipment. This has led to the need for simple, affordable, and portable solutions that can be used by individuals and small organizations.

This project leverages the Arduino microcontroller, which is widely known for its simplicity, flexibility, and cost-effectiveness, along with a TDS sensor to create a compact and user-friendly water quality checker. The system is designed to offer real-time measurements, allowing users to instantly assess the safety of their water supply. This background section highlights the significance of water quality, the role of TDS in water assessment, and the need for an affordable testing solution.

1.2 Problem Statement

- Many people lack access to affordable & reliable water quality testing methods.
- Existing methods are expensive, complex, and time-consuming.
- There is a need for a simple, portable, and cost-effective solution.
- This project provides a solution using an Arduino-based TDS sensor system

1.3 Objectives

The primary objectives of this project are:

- To design and develop a user-friendly, portable Drinking Water Quality Checker using Arduino and a TDS sensor, capable of providing real-time TDS measurements.
- To classify water quality into predefined categories (Excellent, Good, Fair, Poor) based on TDS values, enabling users to make informed decisions about water safety.

1.4 Scope and Application

This project focuses on developing a real-time water monitoring device, with possible real-world use in:

- **Domestic Use:** Easily test the quality of drinking water at home to ensure safety.
- **Industrial Use:** Monitor water quality in industries such as food processing, pharmaceuticals, and chemical production.
- **Environmental Monitoring:** Assess the quality of natural water sources (lakes, rivers, wells) to detect pollution.
- **Educational Purposes:** Use as a teaching tool in schools and colleges for water quality analysis.
- **Portable Water Testing Kits:** Integrate into portable kits for field testing and rapid assessment.

1.5 Report Organization

This report is structured as follows:

- **Chapter 2** presents background study and literature review, focusing on relevant theories, existing methods, and current research trends in water quality checkers.
- **Chapter 3** details system analysis and design, including development methodology, requirements, feasibility, and system modeling.
- **Chapter 4** covers implementation and testing procedures, tools used, test cases, and results.
- **Chapter 5** includes conclusion, identified limitations, and suggestions for future enhancements.
- **References** and **Appendices** provide cited sources and screenshots.

BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Access to clean and safe drinking water is a fundamental human right and a key indicator of public health. However, water contamination due to industrial waste, agricultural runoff, and aging infrastructure remains a serious global issue. Technological advancements, especially in microcontroller-based systems and IoT integration, have enabled the development of portable and real-time water quality monitoring solutions.

2.2 Literature Review

2.2.1 Microcontroller-Based Water Quality Monitoring

Because of its low cost, simplicity of programming, and robust community, the Arduino Uno microcontroller has been frequently used in environmental monitoring applications. Arduino-based platforms have been used by researchers like Sharma et al. (2020) to track a number of water parameters, such as pH, turbidity, and temperature. For remote monitoring, these systems can be combined with wireless communication modules and offer real-time data collecting.

2.2.2 TDS Sensor in Water Quality Assessment

Total Dissolved Solids, or TDS, is an important measure of water quality that shows how much dissolved stuff is in the water, like minerals, salts, and other things. If the TDS level is too high, it can change the taste of the water and might even be bad for your health if you drink it over a long time. The World Health Organization says that water with less than 300 parts per million of TDS is very good for drinking. Many studies have used TDS sensors connected to small computers to get accurate readings. For instance, Gupta and Kumar in 2019 made a cheap system using an Arduino board and a TDS sensor that can tell you right away if the water is safe to drink.

2.2.3 Wireless Data Transmission via Bluetooth

Wireless data sending makes monitoring systems easier to use and more convenient. The HC-05 Bluetooth module is often used to send data between an Arduino and a smartphone because it uses little power and is simple to connect with Android apps. Rajalakshmi and others in 2021 made a water monitoring system where water quality information was sent

to a phone through Bluetooth, allowing people to check the data easily and get alerts in real time.

2.2.4 Mobile Application Integration

Mobile apps provide an easy-to-use way to show water quality information. By connecting the HC-05 Bluetooth module with a specially made Android app, users can get immediate TDS readings without dealing with complicated screens. Earlier studies, like the one by Patel et al. (2022), have shown how apps built with MIT App Inventor and Thunkable can talk to Arduino devices using Bluetooth. These apps offer features such as real-time updates, keeping track of past data, and sending warnings when certain limits are reached.

2.3 Research Gap and Relevance to Present Work

While numerous prototypes have been developed for water quality monitoring using Arduino and various sensors, many focus on multi-parameter systems which, although comprehensive, increase cost and complexity. There is a need for more focused, cost-effective designs that cater to a specific parameter like TDS, especially for rural and low-resource areas where water quality is a pressing concern. There is a growing need for systems that:

- Provide real-time data via mobile apps
- Include threshold-based warnings
- Are battery-operated and portable
- Offer easy interpretation through charts or logs

Our project addresses all these problems.

SYSTEM ANALYSIS AND DESIGN**3.1 Development Methodology**

An Agile-based iterative development model was used for the system, which allowed for modular, testable, and user-focused implementation. The process was broken down into five structured phases, each focusing on a core part of the system. These phases made it possible to keep improving the system and easily adjust to changes, like switching from Bluetooth Low Energy (BLE) to Bluetooth Classic Function.

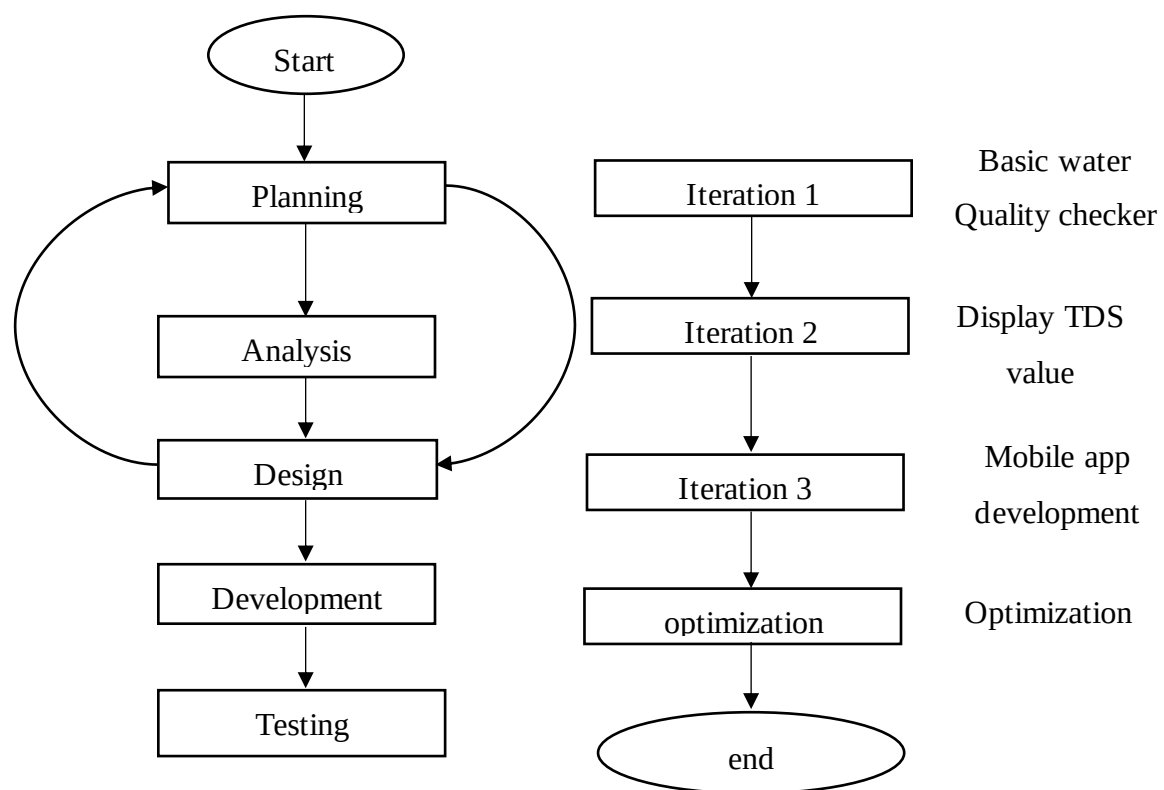


Figure 1: Iterative Development Model

Explanation of the SDLC Phases

- Start: Project initiation and idea conceptualization.
- Planning: Identifying objectives, setting timeline, and gathering required tools/components.
- Analysis: Studying the feasibility and technical requirements, especially how the TDS sensor will work.
- Design: Creating system design for hardware and software components.

- Development: Coding and integrating hardware (TDS sensor, Arduino) with software.
- Testing: Checking sensor accuracy, value display, and overall system reliability

3.1.1 Iteration 1

The initial phase focused on implementing Implementation of the basic water quality checker using a TDS sensor.

3.1.2 Iteration 2

The second phase focused on displaying real-time TDS values on a serial monitor.

3.1.3 Iteration 3

This phase focused on developing of a mobile application to interface with the hardware for remote viewing.

3.1.4 Optimization

This phase was dedicated to Code refinement, improved user interface, and more accurate data representation.

3.1.5 Testing, Deployment and Documentation

The final phase focused on testing the system, creating documentation, and deploying the final version machine ready to use in real-world.

3.2 Requirement Analysis

3.2.1 Hardware

The following are the major hardware requirements of the system:

- **Arduino Uno:** A simple, cost-effective, and widely used microcontroller board. It's ideal for this project as it has sufficient input/output pins for the TDS sensor.
Why ? : Affordable, easy to program, and has a large community for support.
- **TDS Sensor:** Used to measure Total Dissolved Solids in the water.
Why ? : Directly measures the water quality in real-time, and is very easy to interface with Arduino using the analog input pins.
Power Supply: USB cable and 9V DC adapter

3.2.2 Software

The Software requirements of the system are:

- Arduino IDE: The primary software for writing, compiling, and uploading the code to the Arduino board.
- Arduino Libraries
- TDS sensor Library: For interfacing the TDS sensor with Arduino.

3.2.3 Data Processing (Backend)

- Arduino Code: The core functionality for reading TDS sensor data and classifying water quality.

Why? : Arduino will process TDS readings and classify the water quality locally without needing a server or cloud-based backend.

- Algorithm:

Use simple thresholds (e.g., TDS < 150 = Excellent, 150-300 = Good, 300-500 = Fair, >1000 = Unsafe) to classify water quality.(According to World Health Organization)

3.2.4 Mobile App

MIT : A powerful no-code platform for building mobile apps to monitor sensors.

Why?: MIT app inventor allows you to create a mobile app with minimal coding, using a drag-and-drop interface. It can connect to Arduino system via Bluetooth (e.g., using an HC-05 module).

3.3 Feasibility Analysis

3.3.1 Technical Feasibility

The project is feasible due to the low cost and easy availability of hardware like the TDS sensor, Arduino Uno, and HC-05 Bluetooth module. It requires only basic Arduino and Bluetooth programming skills, which are beginner-friendly and well-supported. The components are compatible—TDS sensors use analog pins, and the HC-05 connects via serial. A simple mobile app for real-time data display can be built using MIT App Inventor.

3.3.2 Operational Feasibility

The system measures and displays TDS values to assess water purity, offering a plug-and-play solution that's easy for non-technical users. While the sensor provides basic accuracy,

it may require calibration. A simple mobile app enables Bluetooth connection and data viewing.

3.3.3 Economic Feasibility

The project cost-effective and affordable. Below is the total estimated cost:

Parts	Quantity	Price
Arduino Uno	1	1600
TDS sensor	1	1300
HC-05 Module	1	950
Miscellaneous		600
Total		4450

Table 1: Price evaluation

3.3.4 Schedule Feasibility

The Gantt chart in Table 3.3.4.1 presents a 6-week project timeline with well-structured and overlapping activities. The project begins with research and tool familiarization in Weeks 2 and 3, followed by initial testing and the start of interface design in Week 3. Interface design and integration continue through Weeks 4 and 5, alongside testing and debugging, which extend into Week 6. The final week is dedicated to completing documentation and final testing. This schedule ensures a gradual and organized workflow, allowing sufficient time for development, troubleshooting, and reporting.

Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Research						
Familiarization with Tools						
Testing						
Interface Design and Integration						
Testing and Debugging						
Documentation and Report						

Figure 2: Gantt Chart

3.4 System Design

3.4.1 System Architecture

The system is designed with a modular approach to ensure both scalability and maintainability. It comprises several interconnected components, including TDS sensor, Arduino block, HC-05 Bluetooth module, and android app interface. The TDS sensor senses the TDS level of the sample water and provides a level of TDS value. The Arduino receives the value and sends the data to the android app via a wireless media i.e. HC-05 (a Bluetooth module). The android app, based on the WHO TDS water level label for Drinking purpose, provides an output telling whether it is drinkable or not. This modular design allows for easy updates and integration of future enhancements.

System Diagram:

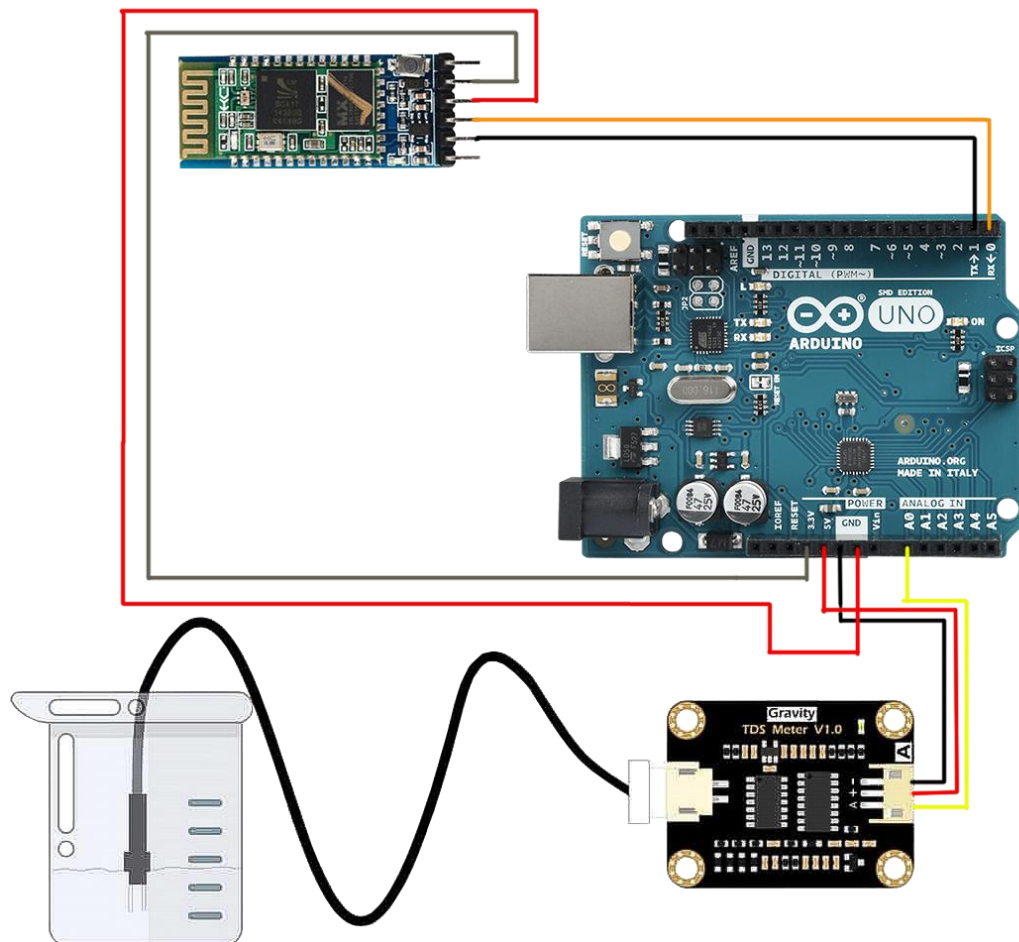


Figure 3: System Diagram

3.4.2 Block Diagram

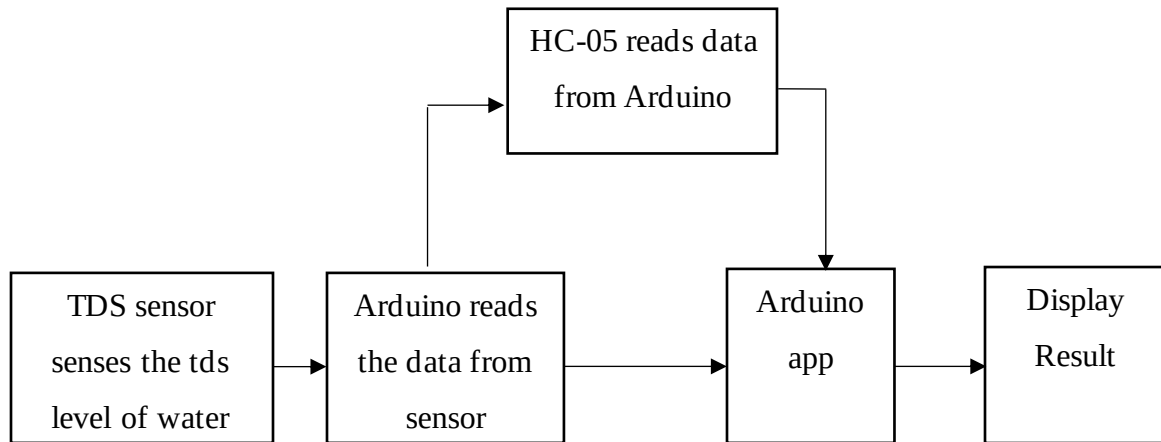


Figure 4: Block Diagram

IMPLEMENTATION AND TESTING**4.1 Data Collection**

The Standard Data Index, according to Indian Bureau Standards, was taken as the distinguishing factor for the each TDS reading given by the system. But we found little difference in comparison between the Indian Bureau Standards index and our sensor readings for the samples. Both Indian Bureau Standards TDS level Index and our Sensor Reading Index are below:

4.1.1 Indian Bureau Standards Recommended TDS level Index

TDS Level (ppm)	Water Quality
0 – 50	Very Low – lacks minerals
50 – 150	Excellent
150 – 300	Good
300 – 500	Acceptable
500 – 1200	Poor – high in salts
Above 1200	Unsafe for drinking

*Table 2: IBS TDS Level Index***4.1.2 Sensor Reading Index**

TDS value (ppm)	Water Quality Description
15 – 50	Pure (Distilled, RO water)
50 – 100	Excellent (Ideal for drinking)
100 – 150	Good (May be contaminated)
150 – 350	Fair (Not recommended for Drinking)
350 or above	Poor (Unsafe for drinking)

Table 3: Sensor Readings

4.1.3 Testing conditions

Parameter	Description
Location	Indoor, outdoor
Device Setup info	Arduino uno, HC-05, TDS sensor
Power Source	USB, Power adapter
Sensor Calibration info	Sensor was calibrated using different samples and IBS index, corresponding data was fetched.

Table 4: Testing conditions

4.1.4 Bluetooth/App Communication Logs

Parameter	Description
Device Pairing Status	Bluetooth connection
Data Transmission Delay	Time lag between Arduino sending & app receiving
Display Accuracy	Is the value on the app the same as on Serial Monitor?

Table 5: Bluetooth-App logs

4.2 Implementation

4.2.1 Tools Used

- Hardware : Arduino Uno, TDS sensor, HC-05 Bluetooth Module, Jumper wires, breadboard
- Software: Arduino IDE, MIT App inventor

4.2.2 Testing

Test Case	Result
Connect sensor to analog pin, observe reading	Values change when water sample changes
Pair phone with HC-05 module	Successful pairing and stable connection
Print TDS values to mobile app	Accurate value within 0–1000 ppm
Sensor to Arduino to Serial Monitor	Live TDS readings visible
Arduino to App via Bluetooth	Values match between Serial Monitor and App
App reconnects after turning Bluetooth off and on	Successful reconnection
Response time from TDS change to app display	Less than 2 seconds
Multiple water sample tests	TDS value reflects water purity correctly

Table 6: Testing

4.2.3 Result Analysis

Water Sample	Observed TDS (ppm)	Water Quality Status
Bottled Water	20-45 ppm	Excellent (safe)
Tap Water	100-150 ppm	Acceptable
River Water	420 ppm	Poor (not suitable to drink)
Rain Water	80 ppm	Good
Pond Water	600 ppm	Unsafe

Table 7: Result analysis

CONCLUSION AND FUTURE RECOMMENDATIONS

5.1 Conclusion

This project is all about successfully developing a portable and cost-effective Drinking Water Quality Checker using an Arduino microcontroller, a TDS sensor. The system accurately measures the Total Dissolved Solids (TDS) in water, classifies water quality (Excellent, Good, Fair, Poor), and displays real-time results. The project addresses the problem of expensive and complex water quality testing methods by providing an affordable and user-friendly solution. The System Development Life Cycle (SDLC) with an Iterative Model was adopted as the project methodology, ensuring continuous improvement and refinement at each stage. The use of MIT app inventor for a mobile application further enhances the user experience, enabling remote monitoring and control. This project has significant applications in domestic, industrial, and environmental water quality monitoring, providing an accessible solution for ensuring safe drinking water.

5.2 Limitations

Despite the successful implementation and favorable results, certain limitations were identified:

- **Limited Water Quality Parameters:** The system only measures TDS (Total Dissolved Solids), which is just one aspect of water quality. It does not measure pH, turbidity, temperature, bacteria, heavy metals, or chemical.
- **Temperature Dependence:** TDS readings are affected by water temperature. Without temperature compensation, the readings may be inaccurate in hot or cold conditions.
- **No Data Storage (in basic version):** The basic version lacks permanent storage of TDS values. Readings are shown live on the phone but are not stored in a database unless Firebase or Google Sheets is added.
- **No Alarm or Notification System:** The system does not automatically alert the user when TDS values exceed safe limits. Manual observation is needed.

- **Not Waterproof or Portable by Design:** The hardware setup on a breadboard is not waterproof or rugged. Not suitable for long-term outdoor deployment without proper casing or enclosure.
- **Power Supply Dependency:** the system typically depends on USB or battery. Power instability or low battery can affect sensor performance or cause disconnection.

5.3 Future recommendation

5.3.1 Add Multiple Water Quality Sensors

- Expand the system to measure additional parameters:
- pH sensor – for acidity/alkalinity detection.
- Turbidity sensor – to detect suspended particles.
- Temperature sensor – to enable temperature compensation for accurate TDS.
- Dissolved Oxygen sensor – for testing aquatic suitability.

5.3.2 Improve Mobile Application

- Make the app more user-friendly and feature-rich:
- Add graphical representation of historical data.
- Include TDS range indicators with color codes (safe/unsafe).
- Allow data export and PDF report generation.
- Add notifications or alarms for unsafe water readings.

5.3.3 Use Better Casing and Waterproof Design

- Make the system durable and field-ready:
- Enclose the circuit in a **waterproof and dustproof box**.
- Use **waterproof connectors and probes**.
- Consider making it **battery-powered** for outdoor use.

5.3.4 Develop a Scalable Version for Communities

- Make it useful beyond individual use:
- Scale it for **municipal water testing, schools, or rural areas**.
- Develop a **dashboard** to compare multiple sensors in different locations.

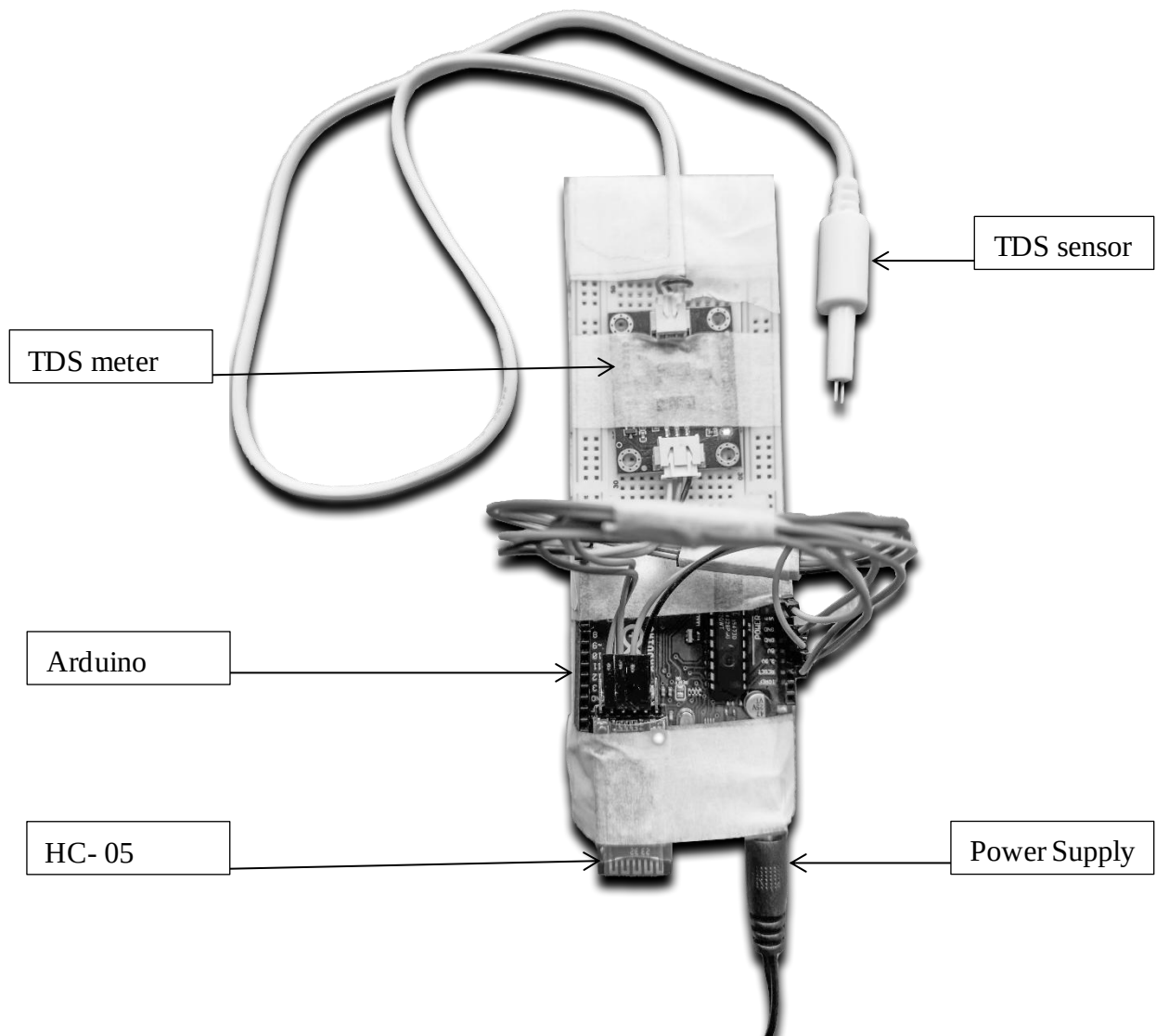
REFERENCES

- [1] N. Luvhimbi, T. G. Tshitangano, J. T. Mabunda, F. C. Olaniyi, and J. N. Edokpayi, “Water quality assessment and evaluation of human health risk of drinking water from source to point of use at Thulamela municipality, Limpopo Province,” *Scientific Reports*, vol. 12, no. 1, p. 6059, Apr. 2022, doi: 10.1038/s41598-022-10092-4.
- [2] M. Rahman, C. Bapery, M. J. Hossain, Z. Hassan, G. M. J. Hossain, and M. Islam, “Internet of Things (IoT) based water quality monitoring system,” *Internet of Things*, vol. 2, no. 4, [Online]. Available: [Publisher details incomplete].
- [3] R. Roy, “An introduction to water quality analysis,” *ESSENCE Int. J. Env. Rehab. Conserv.*, vol. 9, no. 2, pp. 94–100, Dec. 2018, doi: 10.31786/09756272.18.9.2.214.
- [4] Y. Chen and D. Han, “Water quality monitoring in smart city: A pilot project,” *Automation in Construction*, vol. 89, pp. 307–316, May 2018, doi: 10.1016/j.autcon.2018.02.008.
- [5] M. T. R. et al., “Water quality level estimation using IoT sensors and probabilistic machine learning model,” *Hydrology Research*, vol. 55, no. 7, pp. 775–789, Jul. 2024, doi: 10.2166/nh.2024.048.
- [6] M. A. Hannan, M. J. A. Aziz, A. Hussain, S. A. Samad, and A. M. H. Amran, “Smart monitoring of water quality parameters using IoT and cloud computing,” *IEEE Access*, vol. 8, pp. 157497–157513, 2020, doi: 10.1109/ACCESS.2020.3019606.
- [7] S. Mehmood, R. U. Rehman, M. A. Shah, and M. F. Siddiqui, “A low-cost IoT framework for water quality monitoring in smart cities,” in *Proc. 2019 Int. Conf. on Frontiers of Information Technology (FIT)*, Islamabad, Pakistan, 2019, pp. 79–84, doi: 10.1109/FIT47737.2019.00020.
- [8] P. N. Palanisamy, D. Sathishkumar, and R. Sivakumar, “Physico-chemical parameters for testing of water – A review,” *Int. J. Environ. Sci.*, vol. 1, no. 7, pp. 119–124, 2011.
- [9] T. Ahmed, A. Alam, M. N. Islam, M. S. Rahman, and M. S. Uddin, “Design and implementation of a smart water quality monitoring system using Arduino and cloud platform,” in *Proc. 2021 4th Int. Conf. on Electrical and Telecommunication Engineering (ICEAT)*, Dhaka, Bangladesh, 2021, pp. 1–5, doi: 10.1109/ICEAT54917.2021.9689956.

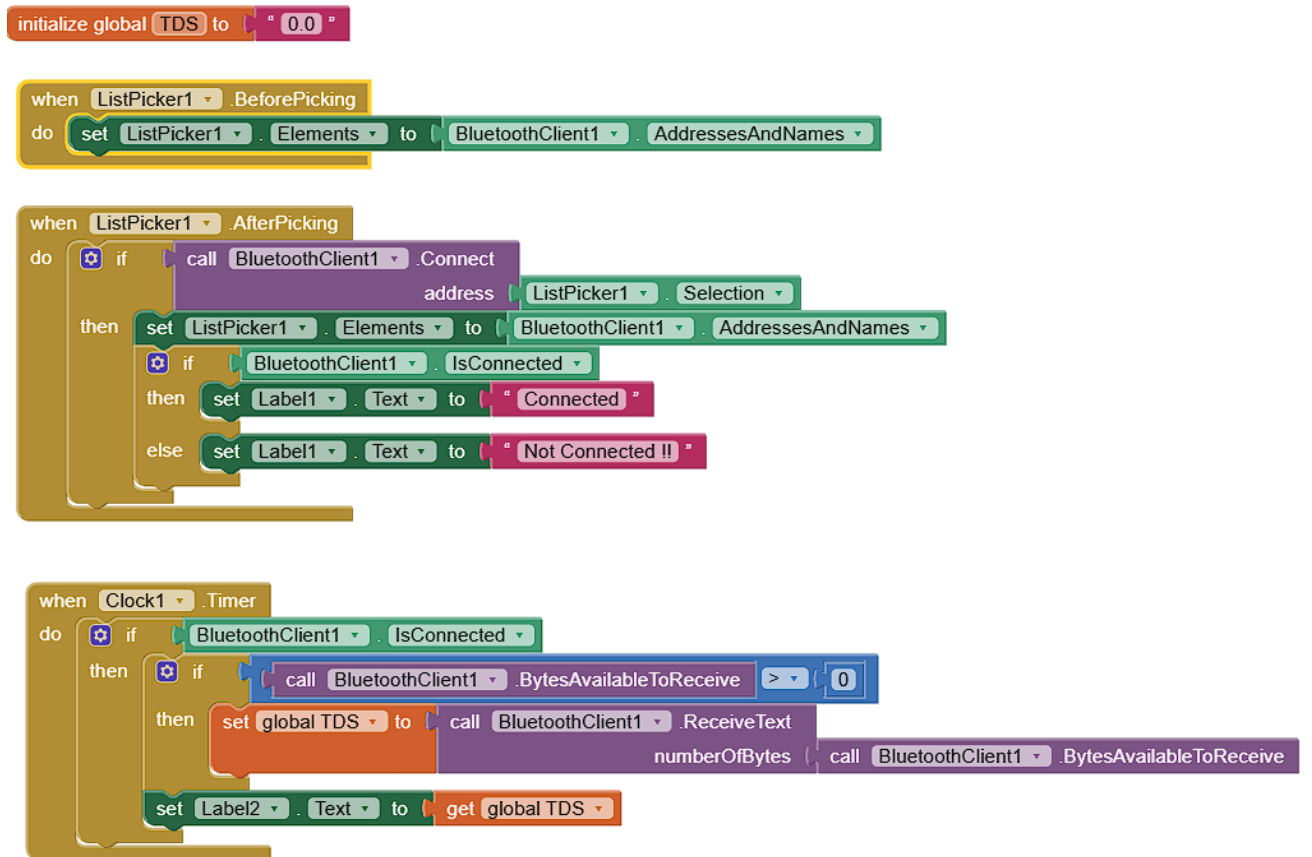
- [10] M. Srivastava and R. Mishra, “Water quality monitoring system using IoT and machine learning,” *Journal of Ambient Intelligence and Humanized Computing*, vol. 13, pp. 1419–1432, 2022, doi: 10.1007/s12652-021-03078-7.
- [11] A. Shinde, P. Patil, and S. Patil, “IoT based water quality monitoring system using Arduino,” in *Proc. 2018 2nd Int. Conf. on Trends in Electronics and Informatics (ICOEI)*, Tirunelveli, India, 2018, pp. 921–925, doi: 10.1109/ICOEI.2018.8553909.

APPENDICES

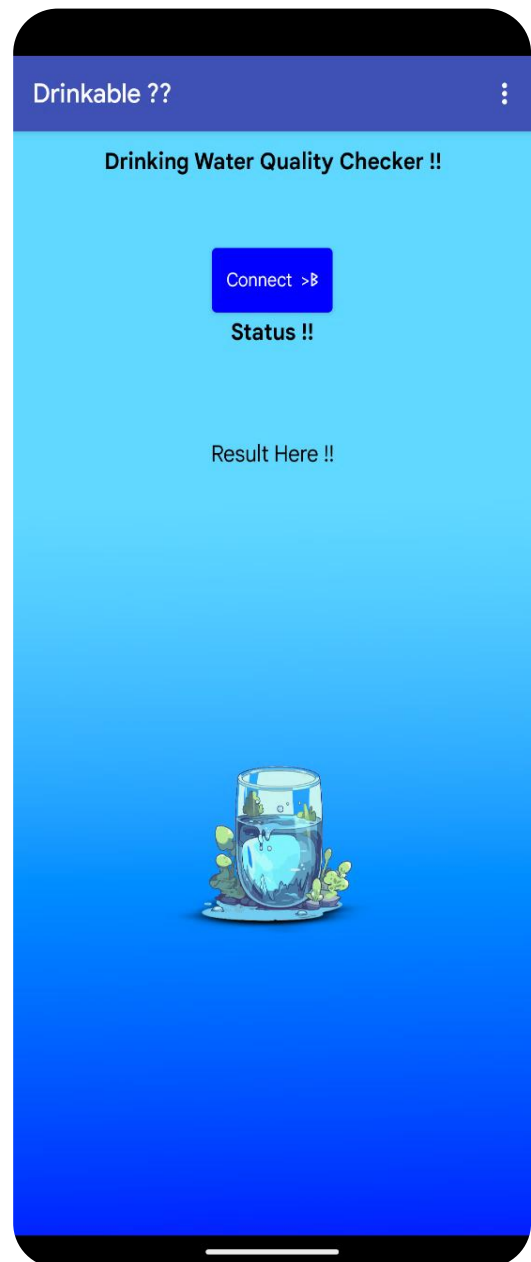
Appendix A: Final System

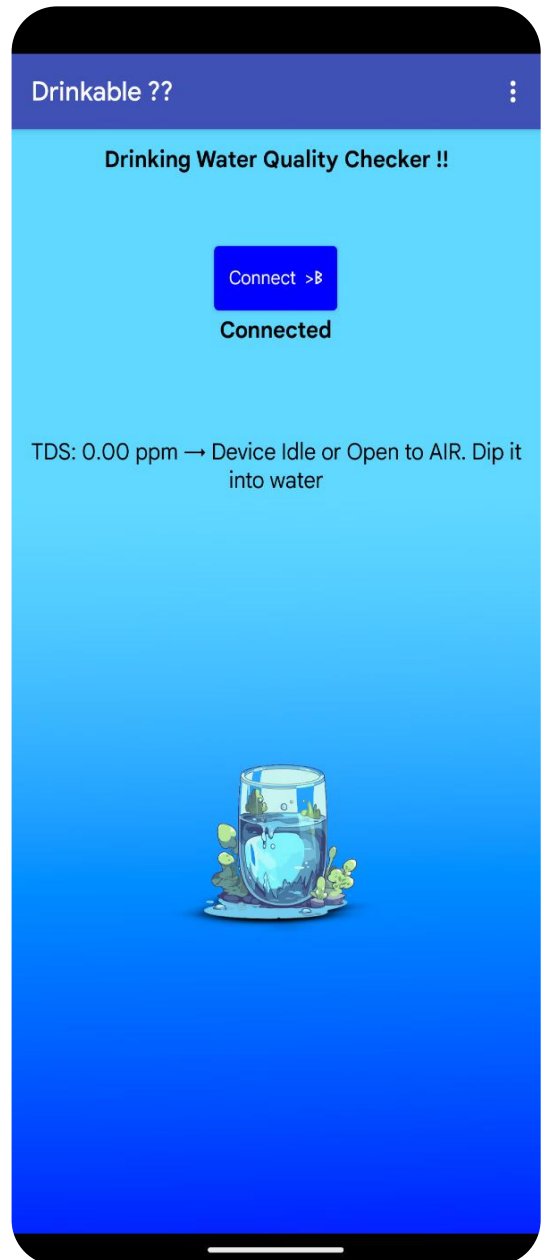
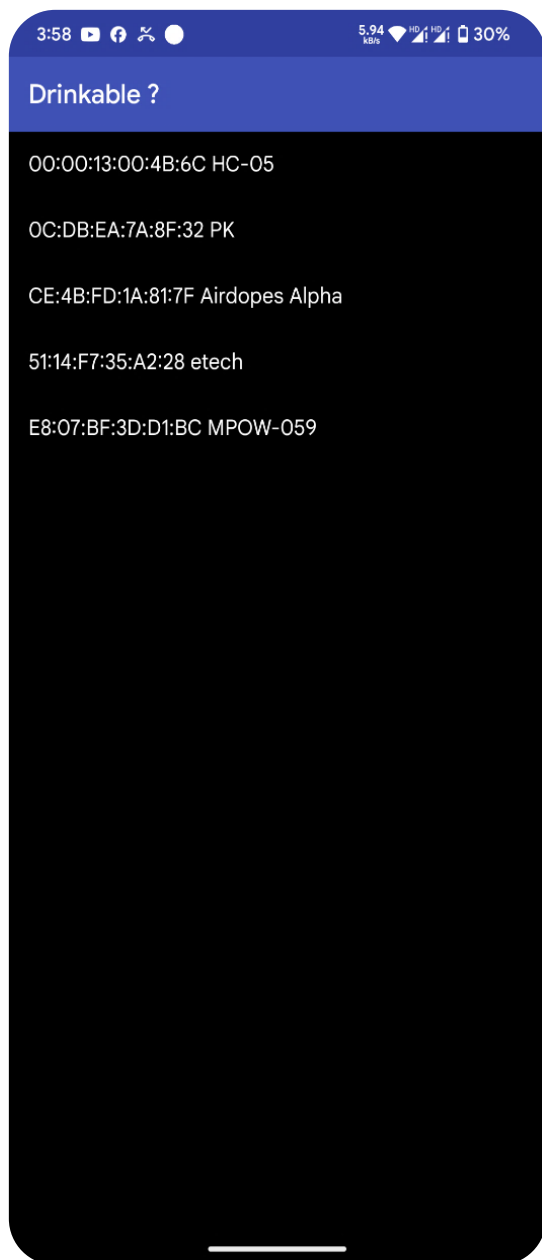


Appendix B: App Logic



Appendix C: App Screenshot





Appendix D: Code



```
1  #include <SoftwareSerial.h>
2
3  #define TdsSensorPin A0
4  SoftwareSerial bluetooth(10, 11); // RX, TX
5
6  float analogValue = 0;
7  float voltage = 0;
8  float tdsValue = 0;
9
10 void setup() {
11     Serial.begin(9600);           // Serial Monitor
12     bluetooth.begin(9600);        // HC-05 Baud Rate
13     pinMode(TdsSensorPin, INPUT);
14 }
15
16 void loop() {
17     analogValue = analogRead(TdsSensorPin);
18     voltage = analogValue * (5.0 / 1024.0);
19     tdsValue = (133.42 * voltage * voltage * voltage
20               - 255.86 * voltage * voltage
21               + 857.39 * voltage) * 0.5; // TDS formula from sensor datasheet
22
23
24     if(tdsValue <= 15){
25         Serial.print("TDS: ");
26         Serial.print(tdsValue, 2);
27         Serial.println(" ppm → Device Idle. Dip it into water");
28
29         bluetooth.print("TDS: ");
30         bluetooth.print(tdsValue, 2);
31         bluetooth.println(" ppm");
32
33         delay(1000); // Send every 1 second
34
35     }
```

Code contid...



```
1  else if (tdsValue > 15 && tdsValue <= 50){
2      Serial.print("TDS: ");
3      Serial.print(tdsValue, 2);
4      Serial.println(" ppm → Pure, RO water");
5
6      bluetooth.print("TDS: ");
7      bluetooth.print(tdsValue, 2);
8      bluetooth.println(" ppm");
9
10     delay(1000); // Send every 1 second
11 }
12 else if (tdsValue > 50 && tdsValue <= 100){
13     Serial.print("TDS: ");
14     Serial.print(tdsValue, 2);
15     Serial.println(" ppm → Excellent, Drinkable");
16
17     bluetooth.print("TDS: ");
18     bluetooth.print(tdsValue, 2);
19     bluetooth.println(" ppm");
20
21     delay(1000); // Send every 1 second
22 }
23 else if (tdsValue > 100 && tdsValue <= 150){
24     Serial.print("TDS: ");
25     Serial.print(tdsValue, 2);
26     Serial.println(" ppm → Good(may be contaminated)");
27
28     bluetooth.print("TDS: ");
29     bluetooth.print(tdsValue, 2);
30     bluetooth.println(" ppm");
31
32     delay(1000); // Send every 1 second
33 }
34 else if (tdsValue > 150 && tdsValue <= 350){
35     Serial.print("TDS: ");
36     Serial.print(tdsValue, 2);
37     Serial.println(" ppm → Not Recommended!!");
38
39     bluetooth.print("TDS: ");
40     bluetooth.print(tdsValue, 2);
41     bluetooth.println(" ppm");
42
43     delay(1000); // Send every 1 second
44 }
45 else if (tdsValue > 350){
46     Serial.print("TDS: ");
47     Serial.print(tdsValue, 2);
48     Serial.println(" ppm → Poor, Unsafe");
49
50     bluetooth.print("TDS: ");
51     bluetooth.print(tdsValue, 2);
52     bluetooth.println(" ppm");
53
54     delay(1000); // Send every 1 second
55 }
56
57 }
```